

Assignment 2

Multi-Layer Perceptron (MLP) Classifier on the Wine Dataset

1. Aim

To implement and evaluate Multi-Layer Perceptron (MLP) classifiers on the Wine dataset and compare their performance under two different train-test split scenarios.

2. Problem Statement

Build MLP classifiers for multiclass wine classification, compare two network architectures, and analyze how the proportion of training and testing data affects measured performance.

3. Dataset

The scikit-learn Wine dataset contains 178 samples, 13 numerical features, and 3 target classes. It is a multiclass classification problem.

4. Methodology

Load the dataset; create stratified 70:30 and 90:10 splits; standardize features using only training data; train MLP1 and MLP2; evaluate accuracy, weighted F1, precision, recall and 10-fold CV F1; visualize confusion matrices and loss curves; compare results.

5. Model Architecture

MLP1: Input → Dense(100, ReLU) → Dense(100, ReLU) → Output. MLP2: Input → Dense(50, ReLU) → Output. Both use Adam optimization, learning_rate_init=0.001, max_iter=2000 and random_state=42.

6. Important Implementation Code

```
wine = datasets.load_wine()
X, y = wine.data, wine.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.30, random_state=42, stratify=y
)

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

model = MLPClassifier(
    hidden_layer_sizes=(100, 100),
    max_iter=2000,
    learning_rate_init=0.001,
    random_state=42
)

model.fit(X_train, y_train)
y_pred = model.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
```

7. Results

Scenario	Model	Accuracy	F1 Score	Precision	Recall	10-Fold CV F1	Iterations
1	MLP1 (100,100)	0.9815	0.9815	0.9815	0.9815	0.9923	117
1	MLP2 (50)	0.9815	0.9815	0.9815	0.9815	0.9841	377
2	MLP1 (100,100)	1.0000	1.0000	1.0000	1.0000	0.9938	123

Scenario	Model	Accuracy	F1 Score	Precision	Recall	10-Fold CV F1	Iterations
2	MLP2 (50)	1.0000	1.0000	1.0000	1.0000	0.9685	398

8. Result Analysis

Both architectures performed very strongly after standardization. In the 70:30 split, both achieved 98.15% accuracy. In the 90:10 split, both achieved 100% accuracy on the small test set. Cross-validation provides a more robust estimate and should be considered alongside the single test split.

The original notebook was corrected because its cross-validation code referenced `X_train` and `y_train` outside the function. The revised version uses the correct scenario-specific training data. Feature standardization is also applied because MLP optimization is sensitive to feature scale.

9. Conclusion

The experiment successfully demonstrates MLP-based multiclass classification on the Wine dataset. Architecture and train-test split affect measured performance, while cross-validation helps assess generalization more reliably.

10. Full Code

The complete corrected code is provided in `mlp_classifier_on_wine_dataset.ipynb` in the assignment folder.